NORTHERN ARIZONA UNIVERSITY
Steve Sanghi College of Engineering

Memorandum

MA3: Advanced Menu and PWM

to: course staff

from: project team

context: undergraduate engineering project

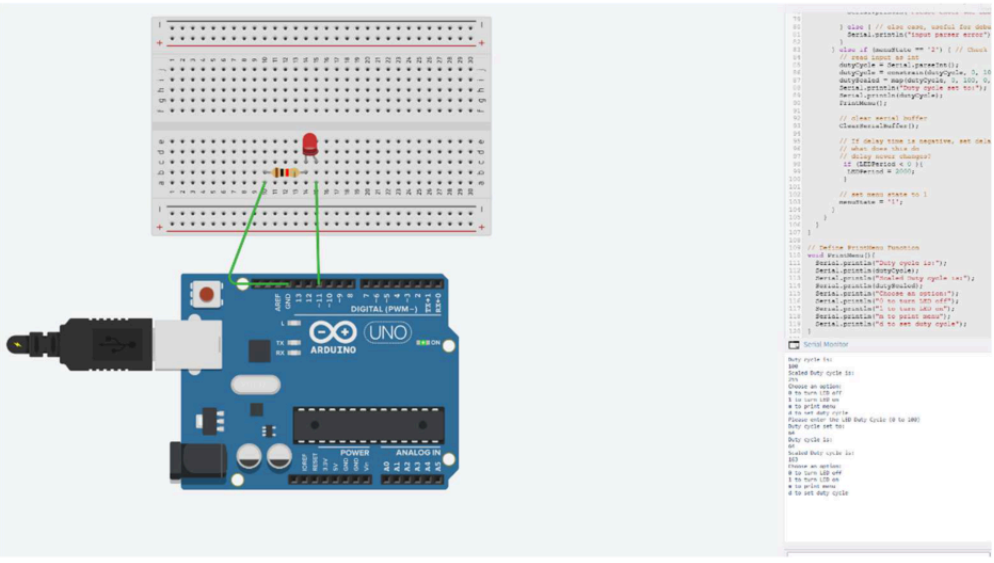
Re: MA3; Advanced Menu and PWM

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1. ***Why is it beneficial to control the LED (or read sensor data and control the Motorcycle) at a high frequency?***
    a. It is beneficial to control the LED or read sensor data at a high frequency because in some cases, actuation and data reading must happen as soon as possible. For example, in controlling the motorcycle's balance, it makes sense to read gyroscopic data and act on it as often as possible.

2. ***Why is it appropriate to print data to the serial monitor and read serial inputs at a slow frequency?***
    a. It is appropriate to print data to the serial monitor and read serial inputs at a slow frequency because the monitor and inputs are both controlled by humans. It is unrealistic to check for human inputs 100 times per second, as a human can't input or parse through data at as high a frequency, compared to the Arduino. Additionally, it can clog up the console, and interfere with time-sensitive actuation or sensing.

3. ***Please explain how can you modify this code to set additional program parameters (for example, the rate of the two timed statements)***
    a. I could modify this code to set additional program parameters by adding an option in the menu, reading the corresponding input, and modifying the time period variables. This would require the time period variables to be initialized as integers, rather than defined with #define.



Snapshot of menu navigation and circuit setup in TinkerCAD.